Question 1 : Explain the difference between Absolute, Relative, and Mixed Cell Referencing in Excel with examples.
Answer:

🔹 1. Relative Cell Reference

Notation: A1

Behavior: Adjusts both row and column when copied.

Example:

Formula in C1: =A1 + B1

Copy to C2 → becomes =A2 + B2

Use Case: Best for repetitive calculations across rows or columns (e.g., summing values row by row).

🔹 2. Absolute Cell Reference

Notation: A1

Behavior: Both row and column remain fixed when copied.

Example:

Formula in C1: =A1 * B5

Copy to C2 → becomes =A2 * B5

Use Case: Useful when multiplying values by a constant (e.g., tax rate stored in B5).

🔹 3. Mixed Cell Reference

Notation: $A1 or A$1

Behavior: Locks either the row or the column, but not both.

Examples:

$A1: Column A fixed, row changes when copied.

Formula in C1: =$A1 + B1

Copy to C2 → =$A2 + B2

A$1: Row 1 fixed, column changes when copied.

Formula in C1: =A$1 + B1

Copy to D1 → =B$1 + C1

Use Case: Perfect for structured tables (like multiplication tables) where one dimension must remain constant.

Question 2 : What is a Macro in Excel? How does it help in automation?
Answer:

- What is a Macro in Excel?

A Macro in Excel is a sequence of instructions or actions that you record or write (using VBA – Visual Basic for Applications) to automate repetitive tasks.

Think of it as a "shortcut program" inside Excel.

Instead of manually performing steps again and again, you can record them once and run the macro to repeat the process instantly.

- How Does a Macro Help in Automation?

Macros save time and reduce errors by automating tasks. Here's how:

Repetition Made Easy

Example: Formatting hundreds of rows (font, color, borders) can be automated with a single macro instead of doing it manually.

Complex Calculations

Example: A macro can apply formulas across multiple sheets, consolidate data, and generate reports automatically.

Data Handling

Example: Importing data from external files, cleaning it (removing duplicates, adjusting formats), and preparing it for analysis.

Consistency

Ensures the same steps are applied every time, reducing human error.

Productivity Boost

Tasks that take hours manually can be completed in seconds with a macro.

Question 3 : What are Text Functions in Excel? Mention any five with examples.
Answer:

🔹 What are Text Functions in Excel?

Text functions are built-in formulas in Excel that help you manipulate, format, and analyze text strings. They are especially useful when working with names, codes, or any non-numeric data.

🔹 Five Common Text Functions with Examples

LEFT Function

Extracts characters from the left side of a text string.

Example: =LEFT("Excel", 3) → Result: "Exc"

RIGHT Function

Extracts characters from the right side of a text string.

Example: =RIGHT("Excel", 2) → Result: "el"

LEN Function

Returns the number of characters in a text string (including spaces).

Example: =LEN("Excel") → Result: 5

CONCATENATE / CONCAT Function

Joins two or more text strings together.

Example: =CONCAT("Data"," ","Science") → Result: "Data Science"

UPPER / LOWER / PROPER Functions

Change the case of text.

Example:

=UPPER("excel") → Result: "EXCEL"

=LOWER("EXCEL") → Result: "excel"

=PROPER("data science") → Result: "Data Science"

Question 4 : What is the use of Scenario Manager in decision making?
Answer:

🔹 What is Scenario Manager in Excel?

Scenario Manager is a tool in Excel (found under the **Data** → **What-If Analysis** menu) that allows you to create and compare different sets of input values (scenarios) to see how they affect the outcome of formulas.

It's part of Excel's **What-If Analysis** features and is especially useful for decision-making when you want to evaluate multiple possibilities without changing your worksheet repeatedly.

🔹 Use of Scenario Manager in Decision Making

Scenario Manager helps in decision making by allowing you to:

1. **Test Multiple Possibilities**
   - Example: A company can create scenarios for *Best Case*, *Worst Case*, and *Most Likely Case* sales forecasts to see how profits change.
2. **Compare Outcomes Quickly**
   - Instead of manually changing inputs, you can switch between scenarios instantly and view the impact on results.
3. **Plan Strategically**
   - Managers can evaluate different strategies (e.g., increasing prices vs. reducing costs) and choose the most profitable option.
4. **Risk Analysis**
   - Helps in understanding the impact of uncertain factors (like market demand or raw material costs) on business performance.
5. **Presentation of Options**
   - Useful for presenting alternative plans to stakeholders with clear data-driven outcomes.

Question 5 : Define the purpose of VLOOKUP and HLOOKUP. How are they different from XLOOKUP? Which among XLOOKUP and INDEX-MATCH is best while usage?

Answer:
🔹 Purpose of VLOOKUP and HLOOKUP
VLOOKUP (Vertical Lookup):
Searches for a value in the first column of a table and returns a value from another column in the same row.

Example: =VLOOKUP(101, A2:D10, 2, FALSE) → Finds employee ID 101 in column A and returns the corresponding name from column B.

HLOOKUP (Horizontal Lookup):
Searches for a value in the first row of a table and returns a value from another row in the same column.

Example: =HLOOKUP("Jan", A1:H5, 3, FALSE) → Finds January in row 1 and returns the value from row 3 of the same column.

🔹 Difference from XLOOKUP
XLOOKUP is a newer, more powerful function that replaces both VLOOKUP and HLOOKUP.

Key differences:

Direction: XLOOKUP can search both vertically and horizontally in one function.

Flexibility: Unlike VLOOKUP, it doesn't require the lookup value to be in the first column/row.

Exact Match Default: XLOOKUP defaults to exact match, while VLOOKUP/HLOOKUP default to approximate match unless specified.

Return Multiple Values: XLOOKUP can return entire arrays (multiple columns/rows), which VLOOKUP/HLOOKUP cannot.

Error Handling: XLOOKUP has a built-in option to return a custom message if the lookup value isn't found.

Example:
=XLOOKUP(101, A2:A10, B2:B10, "Not Found") → Looks for employee ID 101 in column A and returns the name from column B, or "Not Found" if missing.

🔹 XLOOKUP vs. INDEX-MATCH
INDEX-MATCH is a combination of two functions:

INDEX returns the value from a specific row/column.

MATCH finds the position of a value in a range.

Example: =INDEX(B2:B10, MATCH(101, A2:A10, 0)) → Finds employee ID 101 in column A and returns the corresponding name from column B.

Comparison:
XLOOKUP Advantages: Easier to write, more readable, built-in error handling, can return multiple results.

INDEX-MATCH Advantages: More flexible in older versions of Excel (before XLOOKUP existed), can handle complex lookups with multiple criteria.

Best Choice Today:

If you're using modern Excel (Office 365 or Excel 2021+), XLOOKUP is best because it's simpler and more powerful.

If you're working in older versions, INDEX-MATCH is the better option since XLOOKUP isn't available.

Question 6 : Create a dataset of 8 employees with joining dates. Use the dummy dataset to calculate their experience in years and months.
Answer:

EmployeeID	EmployeeName	JoiningDate	Experience (Years, Months)
1	John Doe	2018-01-01	5 years, 9 months
2	Jane Smith	2020-06-01	3 years, 10 months
3	Bob Johnson	2015-03-01	8 years, 7 months
4	Maria Rodriguez	2019-09-01	4 years, 1 month
5	David Lee	2016-11-01	6 years, 11 months
6	Emily Chen	2021-02-01	2 years, 8 months
7	Michael Brown	2017-08-01	5 years, 8 months
8	Sophia Patel	2022-05-01	**1 year, 5 months**

To calculate experience, we'll use the following SQL query (assuming the table name is Employees):

```
SELECT
   EmployeeID,
   EmployeeName,
   JoiningDate,
   TIMESTAMPDIFF(YEAR, JoiningDate, CURDATE()) AS Years,
   TIMESTAMPDIFF(MONTH, JoiningDate, CURDATE()) % 12 AS Months
FROM Employees;
```

Question 7 : You are provided with a dataset containing details of 100 road accidents.Design a Road Accident Analysis Dashboard using Pivot Tables and Slicers in Excel.
Dataset Link

Answer:

1. Data Preparation

- Ensure the dataset contains relevant fields such as:
    - Date of Accident
    - Location
    - Vehicle Type
    - Cause of Accident
    - Severity (Minor, Major, Fatal)
    - Time of Day
    - Number of Casualties
- Clean the data (remove blanks, ensure consistent categories).

2. Create Pivot Tables

Generate multiple Pivot Tables to summarize accident data:

- **Accidents by Location** → Count of accidents per location.
- **Accidents by Vehicle Type** → Count of accidents per vehicle type.
- **Accidents by Cause** → Count of accidents per cause.
- **Accidents by Severity** → Distribution of minor, major, and fatal accidents.
- **Accidents by Time of Day** → Trend of accidents across morning, afternoon, evening, night.

3. Add Pivot Charts

Convert Pivot Tables into charts for visualization:

- Bar Chart → Accidents by Location.
- Pie Chart → Vehicle Type distribution.
- Column Chart → Causes of accidents.
- Stacked Column Chart → Severity distribution.
- Line Chart → Accident trend over time.

4. Insert Slicers

Add slicers for interactivity:

- **Location**, **Vehicle Type**, **Severity**, **Date**.
- Slicers allow filtering across all Pivot Tables simultaneously, making the dashboard dynamic.

5. Dashboard Layout

- Place all charts on a single worksheet.
- Position slicers at the top or side for easy filtering.
- Use consistent colors (e.g., red for fatal accidents, yellow for minor).
- Add a clear title: *"Road Accident Analysis Dashboard"*.

6. Insights from Dashboard

- Identify **accident-prone locations**.
- Determine which **vehicle types** are most involved.
- Highlight **leading causes** (e.g., speeding, drunk driving).
- Show **severity distribution** (minor vs. fatal).
- Detect **peak accident times** (rush hours, late night).

Final Deliverable

The Road Accident Analysis Dashboard provides stakeholders (traffic police, city planners, safety officers) with an interactive tool to:

- Filter accident data by multiple dimensions.
- Visualize accident hotspots and patterns.
- Make data-driven decisions for road safety improvements.

Question 8 : Create a table of 10 products with stock levels. Use Macros to highlight: ● Low stock (less than 10 units) in red ● Overstock (more than 50 units) in green.
Answer:

Stock Level Highlighting with Macros

We create a table of 10 products with their stock levels, then use VBA Macros to automatically highlight **low stock (<10 units)** in **red** and **overstock (>50 units)** in **green**.

1. Sample Product Table

Product ID	Product Name	Stock Level
P001	Laptop	8
P002	Smartphone	12
P003	Tablet	55
P004	Headphones	6
P005	Keyboard	25
P006	Mouse	52

P007	Monitor	9
P008	Printer	15
P009	Router	4
P010	Smartwatch	60

```
Sub HighlightStockLevels()
   Dim ws As Worksheet
   Dim rng As Range
   Dim cell As Range

   'Set worksheet and range
   Set ws = ThisWorkbook.Sheets("Sheet1")
   Set rng = ws.Range("C2:C11") 'Stock levels column

   'Loop through each cell
   For Each cell In rng
      If IsNumeric(cell.Value) Then
         If cell.Value < 10 Then
            cell.Interior.Color = vbRed
         ElseIf cell.Value > 50 Then
            cell.Interior.Color = vbGreen
         Else
            cell.Interior.Color = xlNone
         End If
      End If
   Next cell
End Sub
```

3. Steps to Implement

1. Enter the product table in Excel (Products in column B, Stock Levels in column C).
2. Press **Alt + F11** to open the VBA editor.
3. Insert a new module and paste the above code.
4. Close the editor and run the macro (**Alt + F8** → **HighlightStockLevels** → **Run**).
5. The macro will automatically highlight:
   - **Red** → Stock less than 10 units.
   - **Green** → Stock more than 50 units.

4. Outcome

- Products with dangerously low stock (e.g., Laptop, Headphones, Router) are highlighted in **red**.
- Products with excess stock (e.g., Tablet, Mouse, Smartwatch) are highlighted in **green**.
- Normal stock levels remain uncolored.

Question 9 : You are given a dataset : Create a drop-down list of product names in a separate cell using Data Validation, and write a formula in the adjacent cell so that when a product is selected from the drop-down, its corresponding price is automatically displayed.
Dataset Link
Answer:

Product Drop-Down with Price Lookup

We are given a dataset of products with their prices. The task is to create a drop-down list of product names in one cell, and display the corresponding price in the adjacent cell automatically.

1. Create the Dataset

Example table (assume it is in range **A2:B11**):

Product Name	Price
Laptop	55000
Smartphone	30000
Tablet	20000
Headphones	2500
Keyboard	1500
Mouse	800
Monitor	12000
Printer	7000
Router	3500
Smartwatch	9000

2. Create Drop-Down List

1. Select a cell (e.g., **D2**) where you want the drop-down.
2. Go to **Data** → **Data Validation**.

3. In **Allow**, choose **List**.
4. In **Source**, select the product names range (**A2:A11**).
5. Now cell **D2** contains a drop-down list of product names.

3. Write Formula to Display Price

In the adjacent cell (e.g., **E2**), enter the formula:

=VLOOKUP(D2, A2:B11, 2, FALSE)

- **D2** → Selected product from drop-down.
- **A2:B11** → Table range containing product names and prices.
- **2** → Column number of the price in the table.
- **FALSE** → Ensures exact match.

4. Outcome

- When a product is selected from the drop-down in **D2**, its corresponding price is automatically displayed in **E2**.
- Example: Selecting *Tablet* shows **20000** in the adjacent cell.

Final Deliverable

This setup provides an interactive way to quickly check product prices:

- **Drop-down list** for product selection.
- **Formula (VLOOKUP)** for automatic price display.
- Useful for inventory management, billing, or product catalogs.

Question 10 : Case Scenario :- Dataset Link You are a data analyst working for a retail firm. You've received a sales dataset that includes customer transactions from various regions and product categories. Your job is to analyze this data, clean inconsistencies, extract insights, and build visual reports.
Task :
1. Apply data validation in the Units Sold column to ensure only values between 1 and 20 are allowed.
2. Highlight rows where Profit > ₹5000
3. Highlight any phone number that is not exactly 10 digits
4. Find the Revenue generated by each product and also label according to revenue generated as "High", "Medium", "Low".
5. Find the profit percentage up to two decimal places.
6. Show Total Revenue and Total Profit by Region.
7. Show Average Units Sold per Product Category.

Answer:

1. Apply Data Validation in Units Sold Column

- Select the **Units Sold** column.
- Go to **Data** → **Data Validation**.
- In **Allow**, choose **Whole Number**.
- Set **Minimum = 1** and **Maximum = 20**.
- This ensures only values between 1 and 20 are allowed.

2. Highlight Rows where Profit > ₹5000

- Select the dataset.
- Go to **Conditional Formatting** → **New Rule** → **Use a formula**.
- Enter formula (assuming Profit is in column F):

=$F2>5000
Set formatting to **fill color (e.g., light green)**.
Entire row will be highlighted if Profit > ₹5000.

3. Highlight Phone Numbers Not Exactly 10 Digits

- Select the Phone Number column.
- Conditional Formatting → New Rule → Use a formula.
- Formula (assuming Phone Number is in column G):

=LEN(G2)<>10
Apply formatting (e.g., red fill).
Any number not 10 digits will be highlighted.

4. Find Revenue Generated by Each Product and Label

- Formula for Revenue (assuming Units Sold in column C and Price in column D):

=C2*D2
To categorize Revenue:
=IF(E2>50000,"High",IF(E2>=20000,"Medium","Low"))

This labels each product as **High**, **Medium**, or **Low** revenue.

5. Find Profit Percentage (Two Decimal Places)

- Formula (assuming Profit in column F and Revenue in column E):

=ROUND((F2/E2)*100,2)

Displays profit percentage up to two decimals.

6. Show Total Revenue and Total Profit by Region

- Insert a **Pivot Table**.
- Rows → Region.
- Values → Sum of Revenue, Sum of Profit.
- This shows total revenue and profit per region.

7. Show Average Units Sold per Product Category

- Insert another **Pivot Table**.
- Rows → Product Category.
- Values → Average of Units Sold.
- This displays average units sold per category.

Final Deliverable

The cleaned and analyzed dataset provides:

- **Validated Units Sold entries** (1–20 only).
- **Conditional formatting** to highlight high profits and invalid phone numbers.
- **Revenue categorization** into High, Medium, Low.
- **Profit percentage calculations** with precision.
- **Pivot Table reports** showing total revenue/profit by region and average units sold per category.

This ensures the retail firm has a clear, interactive, and accurate view of sales performance across regions and product categories.